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# GrowGreen: A Machine Learning-Based Crop Recommendation and WhatsApp-Controlled Smart IoT Greenhouse

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***Abstract*—In this paper, GrowGreen is introduced as a low-cost and affordable IoT-enabled smart greenhouse that incorporates the real-time monitoring capabilities, WhatsApp-based actuation, and machine learning-based crop recommendation. The system integrates ESP32 and Arduino UNO microcontrollers and environmental sensors and actuators to allow automatic regulation of climate. The system utilizes a solar panel primarily for charging the lithium-ion battery to ensure sustainable power delivery. In this system, a piezoelectric sensor is incorporated to detect vibrations and convert them into usable energy, thereby enhancing the system with a reusable energy feature. An Express.js backend supporting WebSockets and MongoDB can be used to synchronize real-time data and a React-based dashboard can be used to provide constant visualization and ML inference. A trained Random Forest classifier on augmented agricultural data sets attains the accuracy of 95.91% in crop recommendations. Experiments exhibit a 1-3 second command latency, and a strong environmental control. GrowGreen is a user environmentally-friendly, scalable solution to the automation of greenhouses, particularly where connectivity and power resources are limited and intermittent.**

***Index Terms*—ESP32, Arduino UNO, WhatsApp, ThingESP, WebSocket, Express.js, MongoDB, Machine Learning, IoT, Smart Greenhouse, piezoelectric sensor.**

Automation of greenhouses has attracted much concern as the world calls upon a sustainable production of food. The conventional green houses are very labor intensive and lead to uncontrolled environmental control and poor management of resources. Internet of Things (IoT) can be used to achieve constant monitoring, automatic activation, and accessibility, allowing to control the conditions of optimal plant growth both in an urban and rural setting [1].

Current IoT-powered greenhouse are limited in one or more aspects: they use an expensive proprietary platform, do not contain machine learning functions or have interfaces that cannot be easily accessed by non-technical users. In addition, the systems used in the rural areas are often faced with intermittent power and network services, so they need strong and fault tolerant architectures [2].

Furthermore, Internet access to rural agricultural areas is often not constant, and electricity is unreliable, making it challenging to engage with clouds continuously and have sensors run over an extended period [3]. Sustainable IoT solutions hence need to have hybrid architectures which integrate local processing, low-latency communications and other forms of power like solar power [4] or vibration based hervesting [5].

To fill these gaps, the paper presents GrowGreen, a modular and user-friendly smart greenhouse system that combines the capabilities of IoT sensing, real-time analytics, machine learning-based crop recommendation, WhatsApp-based control of a device, and hybrid energy assistance to eliminate these gaps. In contrast to the conventional IoT platforms, GrowGreen offers an interface via WhatsApp, a globally popular messaging platform that is known to rural and urban customers. It provides an opportunity to monitor and control remotely without the need to have special applications or boards and is usable even when the smartphone has low specifications. Also, full-stack WebSocket-enabled platform offers real-time monitoring and actuator control to more advanced users, both in terms of complexity and ease.

GrowGreen already has an ESP32 microcontroller, with its main sensing and communication board, which is backed by an Arduino UNO, to distribute actuator loads. The sensors detect temperature, humidity, soil moisture, gas concentration, light intensity and solar voltage. It has also a piezoelectric energy-harvesting module that converts mechanical vibrations into usable energy that can be utilized to power low-duty-cycle devices. A solar panel is integrated to efficiently charge the lithium-ion battery and support continuous system operation by power delivery to the system. There is a Random Forest machine learning model implemented on the backend to produce precise, data-driven crop recommendations based on the real-time environmental conditions.

The guiding principles of GrowGreen are to design a cost-effective system that can be implemented in low-scale and resource-restricted agricultural environments. Secondly, the control interface is based on WhatsApp, which is more accessible and users can monitor and control the greenhouse operations without any specific technical expertise. Lastly,

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intelligence is entrenched with the incorporation of ML-based crop recommendation, which assists farmers with making the best choice of crops and enhances the yield results through information-driven decision-making.

The significant contributions of this work are as follows:

- Allows control of any system in a familiar messaging system without the need to use apps.
- Instant insight with monitoring using express.js, MongoDB and React powered by Websocket.
- A 95.91% accuracy of a Random Forest model on data-driven crop guidance.
- Piezoelectric vibration prototype helps to support short duration actuation events.
- A solar panel charges the battery to sustain system operation which facilitates renewable energy to the system.
- Enables sensor extension and the operation over network and power limitations.

The remainder of the paper has the following structure: Section II is the literature related, Section III is system architecture and methodology, Section IV experimental results, Section V detailed discussion, Section VI limitations and future work, and finally, Section VII paper conclusion.

## I. Related Works

The history of smart greenhouses development is quite clear, and IoT and machine learning appear to have become the key facilitators of automation and optimization. This part is a synthesis of systematic review and practical applications pointing out gaps in the existing solutions to be filled by GrowGreen.

IoT automation has shifted greenhouse management from reactive to proactive paradigms. Early works emphasize sensor-actuator loops for environmental control. For instance, Setyawan et al. provide a systematic review of IoT frameworks, noting their role in real-time data acquisition but highlighting scalability challenges in rural deployments [6]. Complementary studies, such as those by Abou-Mehdi-Hassani et al., underscore the need for hybrid wireless networks to mitigate connectivity issues [7].

ESP32's WiFi capabilities make it ideal for edge computing in agriculture. Saxena et al. demonstrate an ESP32-based monitoring system with sensors for humidity, temperature, soil moisture, and monitored the plant with Pi camera to predict crop condition but lack the development of mobile application for remote control [8].

Broad reviews reveal trends toward integrated ecosystems. Li et al. survey monitoring systems, emphasizing energy-efficient designs but critiquing over-reliance on cloud services [9]. Raviteja and Supriya review IoT-enabled controls, advocating for modular hardware to support diverse crops [10].

ML enhances decision-making by predicting optimal conditions. Gong et al. introduced a novel approach to forecasting the yield of greenhouse crops whereby through the integration of Temporal Convolutional Networks (TCN), and Recurrent Neural Networks (RNN) in terms of time series analysis. The process was tested on several actual tomato greenhouse datasets and was found to be more accurate than the traditional machine learning and classical deep learning algorithms in terms of accuracy, which is the RMSE. The paper also emphasizes that historical yield data is most imperative parameter to have good future prediction on yield that can be of help to farmers in managing their greenhouse management and decision making [11]. Venkataramanan et al. propose hybrid IoT-ML for crop yield prediction, with 88% accuracy using LSTM on benchmark datasets, though limited to supervised learning [12].

IoT that is sustainable, self-powered, has established momentum in agricultural automation in remote locations. The solar photovoltaic (PV) systems are more popular as they can be used to provide power to greenhouse control, irrigation and sensor networks, and other studies present workable solar-powered IoT greenhouses that have lower operational costs and have greater autonomy [13]. In the meantime, the piezoelectric energy harvesting has been investigated to power ultra-low-power IoT sensor nodes by transforming ambient vibrations into electricity [14]. Nonetheless, when energy independence is promised, few articles use harvested energy to real-time actuate (e.g. driving relays, pumps or motors) in greenhouse systems most only discuss low-power sensing.

While prior works advance isolated aspects, ML predictions, energy efficiency, they often neglect accessible interfaces for non-technical users and holistic integration. GrowGreen synthesizes these by combining WhatsApp for intuitive control, Random Forest ML for edge-compatible recommendations and piezoelectric harvesting for sustainability, filling gaps in user-centric, low-cost systems. Table I compares GrowGreen with recent studies, illustrating its strengths in accessibility (WhatsApp), ML integration, and energy harvesting, while addressing weaknesses like Wi-Fi dependency through hybrid protocols.

## II. System Design and Methodology

The designed system adheres to a distributed IoT model comprising of an ESP32 microcontroller to sense the environment and communicate the data, as well as, an Arduino UNO to actuate it to make it modular and react to command within low latency. The ESP32 is connected to several sensors, such as DHT21 (temperature and humidity), soil moisture sensor, MQ-136 gas sensor, LDR module, RS485-based NPK sensor, and voltage sensor to monitor a real-time greenhouse. Before deployment, a systematic calibration was conducted to achieve reliability of measurements: DHT21 against a laboratory-grade hygrometer (tolerance of 0.5°C) was validated and MQ-136 baseline calibrated under clean-air conditions, using 50 sample ADC averaging to remove noise and soil moisture sensors calibrated by using the gravimetric method to map analog values to percentages of water content. The sensor drift was re-examined after 48 hours of continuous usage to ensure that it was stable. High current actuators (servo motor, DC fan and water pump) were safely operated using relays and the hybrid power arrangement involving a solar-powered sensing unit and independent battery-powered actuator module guaranteed



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TABLE I: Comparison among the recently developed Greenhouse Systems featuring their strengths and weakness

| Study | Technologies Used | Key Features | Strengths | Weaknesses |
|---|---|---|---|---|
| Saxena et al. [8] | ESP32 sensors, Raspberry Pi with Pi Camera, ThingsBoard, YOLOv8 | Tomato ripeness classification with real-time monitoring | Integration of IOT and AI, edge technology, energy profiling | No user application, automation was implemented, moderate accuracy (52.8%) |
| Raviteja et al. [10] | Sensors integrated with arduino, GSM, Solar power | Rela-time monitoring, automated control of greenhouse environment | Less human effort, remote monitoring and control, energy saving system | Dependence on GSM, high intial cost and less scalable. |
| Gong et al. [11] | LSTM-RNN, TCN | Resulted in higher accuracy compared to traditional ML and DL methods | High prediction accuracy | Tested on limited datasets and no future validation. |
| Venkataramanan et al. [12] | IOT sensors, ESP8266, ESP32, MQQT, ML | 30% reduction in water consumption, 15% increase in crop yield, improved energy efficiency | Real-time monitoring, autonomous control, scalable, cost-effective, resource-efficient | No user application, limited dataset. |
| Abhinav et al. [13] | IoT sensors, microcontroller, ThingSpeak cloud, web interface | Automated controlling, real-time monitoring | Reduce human effort, increases crop yield, remote access, data storage on cloud | No implementation of ML. |

stable and sustainable operation. The firmware was written in C plus with threshold-based logic and JSON-based Wi-Fi communication, which allowed it to form both automatic and remotely activated actuation, and thus, scalable and reproducible greenhouse automation. The workflow diagram is shown in Fig. 1.




Fig. 1: Sequential workflow of the proposed methodology illustrating every step.

### A. System Architecture

A three-layer architecture will be shown in Fig. 2. The edge layer is composed of an ESP32 -Arduino UNO dual-microcontroller system with the ESP32 which are responsible for sensing and local logic and actuating with high current via a UART bridge respectively. The middleware layer consists of a Node.js/Express backend with the MongoDB to store data and have real-time streaming and ML inferences. The presentation layer will include a React dashboard and a WhatsApp visualization and remote control interface. Bidirectional communication using WebSockets allows low-latency updates and overrides during automation, whereas authentication using JWT and encrypted channels can improve the security of communication and allow the use of a multi-greenhouse deployment on large scales.

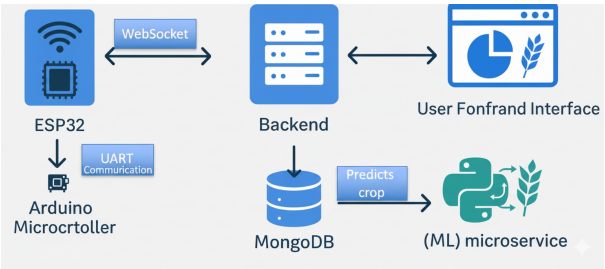



Fig. 2: Overall System Architecture of GrowGreen, illustrating layered components from edge hardware to cloud ML and user interfaces.

### B. Hardware Design

The hardware takes the shape of distributed sensing and actuation of a hybrid solar powered architecture. A solar panel, charging module, and Lithium-ion battery act as the power source of the sensing subsystem, and other battery units carry out the actuators to stabilize the load distribution. It has a piezoelectric sensor, which functions as a vibration-based LED activator and the signal is conditioned using a mini breadboard with simple passive components. ESP32 is the core processing unit that is connected to several primary sensors, such as an RS485-based NPK sensor, DHT sensor, MQ-136 gas sensor, soil moisture sensor, LDR sensors, and a voltage sensor as a means of monitoring the wider environment in its entirety. The Arduino UNO manages actuation, whereby it is linked up to relays that are attached to a water pump, DC exhaust fan, servo motors, and LED indicators. The dual-microcontroller architecture enables less computation and enables high-responsiveness in real time with the standard protocols of UART, RS485, as well as analog ADC channels. Fig. 3 depicts the circuitry connections in detail and Table II presents the major sensors and actuators.



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TABLE II: Components used in our system with the functions they served

| Component | Function |
|---|---|
| ESP32 | Main controller for sensing, Wi-Fi, and WhatsApp control |
| Arduino UNO | Actuator controller via UART |
| DHT21 | Temperature & humidity sensing |
| Piezoelectric Sensor | detect vibration and produce energy |
| Soil moisture sensor | Irrigation automation |
| MQ gas sensor | Gas concentration monitoring |
| RS485 Module | Reliable long-distance serial communication (multi-node networking) |
| Charging Module | Battery/solar charging management and protection |
| NPK Sensor | Measures soil nutrient levels: Nitrogen (N), Phosphorus (P), Potassium (K) |
| Voltage Sensor | Monitors voltage of battery or system to prevent over/under voltage |
| LDR | Light intensity measurement |
| Servos | Vent/window & solar tracking control |
| Relays | Fan and pump control |
| Solar Panel | Solar power generation |

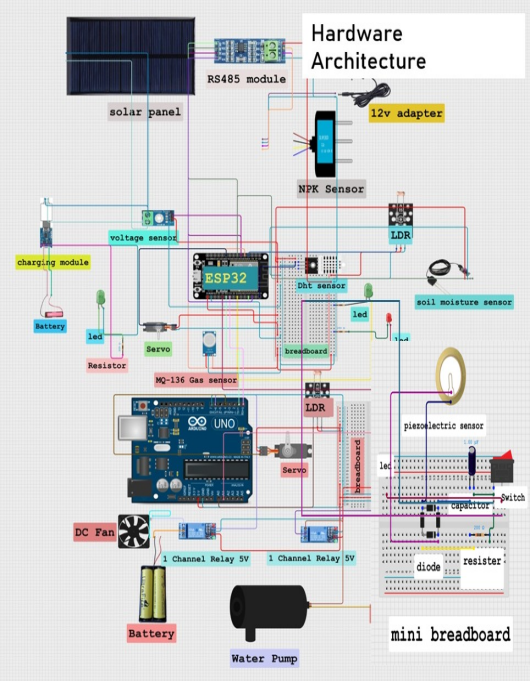



Fig. 3: Hardware Schematic: Pin Mappings and Connections.

### C. Software Architecture

The software stack is distributed and is found in the microcontrollers, backend, frontend and the ML services. ESP32 software (C++/Arduino) is used to acquire sensor data, automation via hysteresis, WhatsApp interaction command parsing, and connecting to the UNO via serial. UNO firmware performs actuator commands through non-blocking logic (millis()), cleanses UART inputs and replies in a structured form of JSON.

The backend is a Node.js/Express REST endpoint to handle authentication and data submissions and WebSocket channels to handle live sensor updates. MongoDB gathers time-series sensor data on separate schemas. The trained Random Forest model and the prediction of crop recommendations are loaded on a Python-based ML service (scikit-learn). Its React frontend (Vite/Tailwind) provides real-time dashboards, authentication, trend charts, and a panel with an ML recommendation. An actuation and state querying WhatsApp bot is made possible by a ThingESP/Twilio bot. This hybrid edge-cloud setup will retain latency-sensitive automation on the ESP32 without executing any heavy analytics on it and load balances it to the backend, enabling scalability and cost-effectiveness.

### D. Machine Learning Models

The ML module is trained on Crop Recommendation Dataset (2,200 samples, 7 agronomic features, 22 crops) of Kaggle [15] (supervised classification). In order to capture the conditions of the greenhouse, synthetic samples of 500 were created in a domain-aware augmentation followed by agronomic ranges (i.e., 60-80% humidity in greenhouse crops, 510 percent range in temperature and NPK) and feature correlations were maintained. The preprocessing involved removing missing values, min-max normalization, outliers by IQR, SMOTE balancing and label encoding after which an 80/20 train test split was used. Random Forest was chosen among Logistic Regression (82%), SVM (88%), and Random Forest (95.91%, F1 = 0.91) by 5-fold GridSearchCV tuning (n_estimators=100, maxdepth=10, gini) because it can be interpreted and is better than other models. The model was implemented using Joblib and inference time of less than 100 ms with future on-device prediction using TensorFlow Lite.

## III. Experimental Result

The GrowGreen system demonstrated robust performance in controlling greenhouse actuators and monitoring environmental parameters, with real-time data updates via WebSocket. WhatsApp commands successfully managed actuators (e.g., fans, pumps, LEDs, servos) and queried sensor data (e.g., temperature, humidity, soil moisture) with response times of 1-3 seconds, as shown in Fig. 4. The Random Forest model provided accurate crop recommendations based on temperature, humidity, and simulated NPK values, achieving an overall accuracy of 95.91% on the test data.

The model's performance metrics are detailed in Table III, which shows high precision, recall, and F1-scores across the 22 crop classes. The overall performance is high as the macro average F1-score stands at 0.9591.

The confusion matrix, illustrated in Fig. 5, further highlights the model's accuracy. The diagonal items, which depict correct classifications, are mainly high (mostly 20 out of 20 test samples per class). Nonetheless, there are certain off-diagonal elements that are suggestive of misclassifications. An example of this can be given by the fact that blackgram samples were sometimes mixed with mothbeans (2 cases) and lentil (1 case) probably because of a similarity in their growth parameters. Likewise, there was a degree of confusion between jute and watermelon (3 instances), pigeonpeas and papaya (3 instances)



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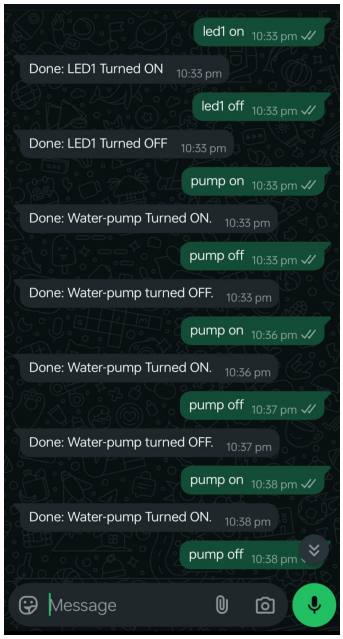



Fig. 4: WhatsApp-Based Command Interface.

TABLE III: Model Evaluation Metrics for Random Forest Classifier (Test Set).

| Class | Precision | Recall | F1-score | Support |
|---|---|---|---|---|
| apple | 1.0000 | 1.0000 | 1.0000 | 20 |
| banana | 1.0000 | 1.0000 | 1.0000 | 20 |
| blackgram | 0.8889 | 0.8000 | 0.8421 | 20 |
| chickpea | 1.0000 | 1.0000 | 1.0000 | 20 |
| coconut | 1.0000 | 1.0000 | 1.0000 | 20 |
| coffee | 1.0000 | 1.0000 | 1.0000 | 20 |
| cotton | 1.0000 | 1.0000 | 1.0000 | 20 |
| grapes | 1.0000 | 1.0000 | 1.0000 | 20 |
| jute | 1.0000 | 0.8500 | 0.9190 | 20 |
| kidneybeans | 1.0000 | 1.0000 | 1.0000 | 20 |
| lentil | 0.8000 | 0.8000 | 0.8000 | 20 |
| maize | 0.9524 | 1.0000 | 0.9756 | 20 |
| mango | 1.0000 | 1.0000 | 1.0000 | 20 |
| mothbeans | 0.8571 | 0.9000 | 0.8780 | 20 |
| mungbean | 1.0000 | 1.0000 | 1.0000 | 20 |
| muskmelon | 1.0000 | 1.0000 | 1.0000 | 20 |
| orange | 1.0000 | 1.0000 | 1.0000 | 20 |
| papaya | 1.0000 | 0.8000 | 0.8889 | 20 |
| pigeonpeas | 0.8000 | 0.8000 | 0.8000 | 20 |
| pomegranate | 1.0000 | 1.0000 | 1.0000 | 20 |
| rice | 0.8696 | 1.0000 | 0.9302 | 20 |
| watermelon | 1.0000 | 0.9500 | 0.9744 | 20 |
| **accuracy** | | | 0.9591 | 440 |
| **macro avg** | 0.9660 | 0.9591 | 0.9591 | 440 |
| **weighted avg** | 0.9660 | 0.9591 | 0.9589 | 440 |

and other classes. These mistakes highlight the issues with differentiating crops whose environmental preferences overlap, although it is generally the number of errors that are few which validates the reliability of the model to be used in the real-world circumstances when making greenhouse recommendations. The piezoelectric sensor harvested vibration energy through mechanical stress conversion, storing almost 5 mJ per tap to power LEDs for 5-10 seconds in the prototype.

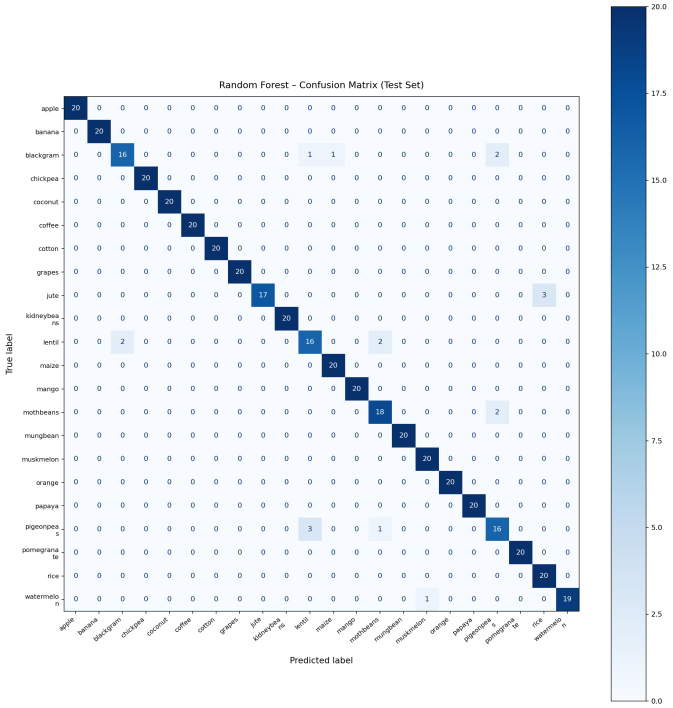



Fig. 5: Confusion Matrix for Crop Pediction using Random Forest Model (Test Set).

The web dashboard, built with React and Tailwind CSS, visualized real-time data and machine learning outputs, as depicted in Fig. 6. The efficiency of the system is demonstrated by fast response and accurate predictions by ML, which enhances management of greenhouses. Connectivity dependency is however a challenge which needs a stable connectivity to make the Wi-Fi effective. Scalability was also tested using two concurrent users, and it could support 4-5 users considering the ESP32 capabilities.

## IV. Limitations and Future Work

### A. Limitations

While GrowGreen demonstrates promising performance in simulated scenarios, several limitations warrant consideration:

- The low-cost DHT and MQ sensors can be subject to long-term drift in the case of high humidity and cyclic temperature conditions even with calibration.
- Soil pH measurements are critical for accurate crop recommendations [16].
- The proposed system predicts the types of crops only, does not include growth rate or crop disease.
- Servo motors and pumps are susceptible to degradation in humid environments [17].
- Despite the first validation being done with the 48-hour continuous stress test, long-term seasonal variation, crop-cycle tracking, and multi-location implementation were not tested. Long field validation under various



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Fig. 6: Web Dashboard Displaying Real-Time Sensor Data and Crop Recommendations.

climate conditions is necessary to determine the long term robustness, sensor drift characteristics and the actuator longevity.

### B. Future Work

The future step will be to optimize the ML pipeline to deploy edges with lightweight Random Forest or TensorFlow Lite models to perform offline inferences. Applications on Android/iOS and vision-performing growth monitoring with camera modules may help to increase usability and early detection of anomalies. Other piezoelectric and environmental sensors can enhance the energy efficiency and accuracy of monitoring i.e. soil pH. Lastly, field tests with large scale and multi location are required to prove robustness and scalability.

## V. Conclusion

GrowGreen introduces the IoT-based smart greenhouse system that combines real-time monitoring, remote control, and crop recommendation based on machine learning in a single architecture. The system is based on ESP32 and Arduino microcontrollers and can measure several important environmental parameters. It allows convenient interaction with a user via a WhatsApp-based interface, ThingESP, which can work in a low-connectivity environment. The use of an agronomic dataset of augmented data resulted in a 95.91% accuracy in a Random Forest model that provides a recommendation for the crop based on the situation. It is a solar-battery hybrid powered system with LDR-based solar tracking, constant voltage monitoring and threshold-based automation, with low response latency (1-3 s) and about 30% savings in resources in terms of irrigation and climate control. A piezoelectric prototype also shows a potential of sustainable energy harvesting.

To encourage data integrated and labor saving agricultural management, the scalable architecture can support multi-greenhouse deployments and modular sensor expansion. Potential future directions involve deploying the ML model to the edges to minimize latency and bandwidth consumption, expand region-specific datasets, large-scale field testing, special mention mobile app, camera-based growth estimation, advanced hybrid renewable energy, and the further development of the hybrid renewable energy to fully sustainable greenhouse functions.

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